

Crowd Monitoring Using Mobile Phones

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Abstract—Crowd density monitoring is crucial to many applications, such as guiding tour and crowd control in commercial environments. Video or image-based solutions are high cost and cannot be applied in low-light environments. For some Radio Frequency (RF) based technologies, some people have to carry certain wireless transceivers, others also need to collect abundant fingerprint. These approaches are high cost and impractical. In this paper, we propose a crowd monitoring approach using mobile phone. Our design of crowd detection adopts clustering methods. Feature sets derive from Wi-Fi signal strength measurements. We use Bluetooth readings analyzing to estimate crowd density. We implement our design on off-the-shelf smartphones and evaluate its performance via extensive experiments in typical real-world scenes. Results of experiment verify the feasibility and the effectiveness of our proposed approach.

keywords—Crowd monitoring; mobile phone; Clustering; Wi-Fi; Bluetooth

I. INTRODUCTION

Crowd monitoring is significant in many fields such as guided tour, human safety monitoring, and crowd control. Monitoring the crowd distribution and density has been a hot and challenging issue in pervasive computing area. Traditional solutions, as video (or image) and RF-based approaches, the monitoring environments are deployed with wireless and other video infrastructures. The cost of infrastructures might be a large problem.

Today, crowd monitoring and density estimating are implemented by video (or image) based recognition technology [1]. However, digital cameras can only cover a small line-of-sight area. In addition, it is difficult to obtain high estimation accuracy when overlap and occlusion exist in crowd. Moreover, video-based method does not work in dark or smoke environments. Furthermore, the privacy of the targets should be concerned.

RF analysis solutions leverage radio signal to locate people or processed crowd. Some of them require the targets to carry certain devices [2], such as sensor nodes, RFID tags, mobile phones, etc. In addition, these solutions need customers of these devices to participate proposed algorithm intentionally. Device-free RF analysis methods employ RSS fingerprints or signal attenuation models for locating objects or estimating crowd density. However, these solutions are difficult to be applied pervasively. The main reason is that they need assign the devices to each person, which is inconvenience for customers. Site survey and pre-installing wireless infrastructures also increase the cost of system. In addition, the radio attenuation model performs poorly due to RF signal are easily affected by

dynamic environments.

Nowadays, with the proliferation of computational ability, wireless communication technologies such as Wi-Fi and Bluetooth has fostered the flourish of smartphone market. Mobile phones can be viewed as an effective interface between users and environments. It provides opportunities to obtain crowd monitoring by low-cost and practical approach.

In this paper, we propose an alternative crowd monitoring method based on mobile phone. Our solution requires people settle or moving in the monitoring environment with their mobile devices sensing on existing wireless infrastructures. The contribution of this work is summarized as follows.

- We propose a novel and low-cost approach to detect crowd and estimate density in indoor spaces using mobile phone. Our solution using clustering methods operating on the feature sets which derived from Wi-Fi signal strength measurements. We use Bluetooth readings analyzing to estimate crowd density.
- We implement our scheme with off-the-shelf mobile phones and evaluate it with extensive real-world experiment.

The rest of this paper is organized as follows. Section II. we review the related work. The proposed approach of crowd detecting and density estimating is elaborated in Section III, followed by the performance evaluation in Section IV. We conclude the work in Section V.

II. RELATED WORK

Recently, indoor localization and crowd monitoring problem has attracted much attention. The solutions can be classified into two categories: video or image-based approach and RF-based solution.

Previous video or image-based work on crowd monitoring have focused on image based recognition technology, these systems use the image from multiple video devices, by the steps of background modeling, changing detection, grouping and event interpretation, they can obtain the distribution and density of crowd. [3], [4] extract information from video sequences of the motion vectors of salient points. Its a method of statistical analysis. [5] In order to estimate the number of moving objects in a crowd scene, the authors provided a clustering technique to cluster feature points tracked across frames.

RF-based solutions rely on Received signal Strength (RSS) analysis. Some of them need the objects carry transceiver equipment, RADAR [6], it is a radio frequency based system

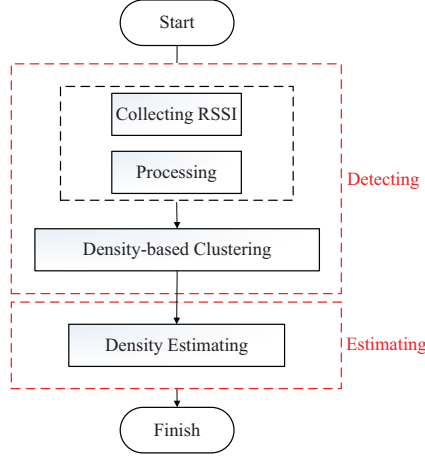


Fig. 1: The work flow of the crowd monitoring approach

for locating and tracking users inside buildings. LANDMARK [7], it collected the RSSI values transmitting from the localized target objects which carry active RFID, [8] utilized the headset to estimate the crowd density. RSS fingerprint based localization object or estimation crowd method [9], [10] has gained much attention in recent years, this kind of scheme divided locating procedure into two phases: off-line training phase and online operation phase. This fingerprint requires significant overhead in construction, in addition to the inflexibility to environment dynamic, this is the main drawback of fingerprint based solution. Signal attenuation models [11] is a common technology for localization, but attenuation models perform poor, due to multipath effect in indoor environments.

III. OUR APPROACH

In this section, the details of our design are presented. We start by an overview of the system design followed by the detail of our solution.

A. System Design

Figure 1 gives an overview vision of the system design. By investigating the characteristics of the Wireless signals strength of nearby Wi-Fi, we discover some useful phenomenon. A key observation as follow: the RSS readings of a same AP keep similarity and stabilization when some people whom carried mobile devices stay or move together in a specific area. our crowd monitor algorithm are implemented in the application server that process the collected wireless signal samples from the users mobile phones. In our solutions, the carried mobile devices of the relevant person need to deploy a sensing service in order to measure the RSS of nearby Wi-Fi. This service was developed on the increasingly popular Android OS.

The working process of our crowd monitoring consists of two phases: crowd detecting and density estimating. During the crowd monitoring phase, In order to infer crowds, we apply clustering techniques operating on the dissimilarity feature between signal vectors. In the crowd density estimating phase, this paper proposed a crowd counting method using

the Bluetooth detection records on the mobile phones. We will elaborate on the detail design of these two phases respectively in the next subsections.

B. Crowd monitoring

The proposed crowd detect method using Wi-Fi signal strength measurements which obtained from the mobile devices carried by users. these system need not installing addition digital devices, such as cameras or sensors, Users just work in offices, consume in shopping malls, or have a rest at coffee shops, walking or sitting. The information of Wi-Fi and Bluetooth signals are collected automatically by their cell phones.

Mobile devices of users will measure the signal strength of nearby Access Points (AP), the signal strength vectors as $s_1, \dots, s_t$. For each time step and device m , a list of vectors S_m in constructed consisting of signal vectors in a period of time. we use the dissimilarity (Euclidean distance) between signal vectors at a specific time as our crowd detect clustering feature. In order to minimize signal noise, we compute the average Wi-Fi RSS value S_{avg} of per AP from the signal vectors S_i ($1 \leq i \leq n$). So the distance function, based on Euclidean distance, for two signal vectors $S_{avg,u}$, $S_{avg,v}$ is defined as:

$$d(S_u, S_v) = ||S_{avg,u} - S_{avg,v}|| \quad (1)$$

Using the dissimilarity measured describe above, We will obtain a dataset D which consist of all the distance estimation from n mobile devices of the users. We incorporate a clustering procedure to partition the collected Wi-Fi signals into cluster. Each cluster corresponds to one crowd.

We use Density-based clustering [12] to cluster different mobile devices. Density-based cluster connect points within a specific distance threshold (Eps) and a minimum number of points (MinPts) within Eps. We adapted DBSCAN [12] algorithm to operate on signal feature. DBSCAN demonstrates its high accuracy and efficiency for this application, which are implemented in WEKA, a popular classification and clustering tool. This algorithm tests whether all the points of dataset D settle within a threshold distance Eps. After set S is created, our algorithm checks if S consists of at least MinPts points. If so, it seeks to merge S and other density joinable clusters, on the contrary, a new cluster is created. Algorithm.1 shows the pseudo-code of the crowd detection.

C. Crowd Density Estimating

Nowadays, mobile phones possess powerful computation and communication capability, and are equipped with Bluetooth sensor for short-range peer to peer communication. In addition, many people set default setting of their Bluetooth transceivers of their mobile phone as the discoverable mode. So we now can analyze Bluetooth readings to estimate crowd density.

Within the reachable communication range of users mobile phone with embedded Bluetooth, users devices continuously sending inquiry to search for the surrounding crowds devices,

Algorithm 1 Density-based crowd detect**Input:** $D, Eps, MinPts$ **Output:** $Crowds$

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1:  $Crowds := \phi$ 
2:  $D :=$  all points of specific time
3: for all points  $p$  of  $D$  do
4:   if  $p \leq Eps$  then
5:      $S \leftarrow p$ 
6:   end if
7:   if  $|S| \geq MinPts$  then
8:     if  $S$  can joined into existing clusters then
9:       combine  $S$  and other clusters
10:    else
11:       $Crowds \leftarrow S$ 
12:    end if
13:  end if
14: end for

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Fig. 2: The experiment scenes

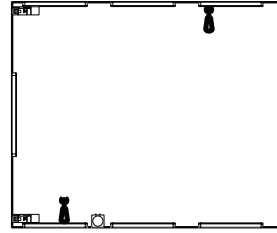


Fig. 3: Floor plan with icons for 2 APs

the protocol for Bluetooth inquiry first receives the Bluetooth Device Address (BDA) of surrounding Bluetooth devices, and then inquiry the names of these devices. BDA is a 48-bit MAC address, and is used for communicating with other devices by sending their BDAs to identify each other. From the reports of BDAs, the crowd density can be estimate by counting the number of BDAs, which read from surrounding Bluetooth devices in the monitor scenario.

IV. EXPERIMENT AND EVALUATION

In this section, we illustrate the implementation and do various experiments to evaluation the performance of our approach in different scenes as shown in Figure 2.

A. Experiment Methodology

We conducted an experiment in an indoor environment to obtain a dataset. The experiment took place in an office building as shown in Figure 3 with 3 to 12 participants. All of them were carried different kinds of smart phones, including SumSung Galaxy S and Google Nexus S. We developed the application on the increasingly popular Android OS. The Wi-Fi signal strength readings are recorded with the frequency of around twice per second.

During the experiment, the number of persons in crowds spanning from 3 to 12, all of participants put their phones into their trouser pockets. They would stay together or moving freely, including meet, merge and divide. The occupied zones

TABLE I: THE SCENES OF THE EXPERIMENT

SA:	1 crowd consist of 3 persons
SB:	1 crowd consist of 12 persons
SC:	2 crowds and each consist of 6 persons
SD:	1 - 3 crowds constructed with changed persons

of crowds lie in a specific subarea of the office building. Stay together was defined as around 1.5m with some variation, and the speed of moving vary at $[0, 1\text{m/s}]$ randomly. Each scene of our experiment was designed to test different situations. We implemented a group of experiments and repeated them some rounds, the obtained signal samples as our test sets. The scenes of the experiment are listed in Table I.

B. Experiment Performance

In order to assess the performance of our approach, we compare the manually estimated results with the estimative result of our experiments. Table II shows the comparison result when the participants were stayed together. The row of the table indicates the crowd level and density classified manually. The column indicates the results classified by our approach. The percentages in the diagonal show the Coincidence Rate (CR) of the manual method and the tests results. To analyze the influence of the person movement to our approach, the same comparison of the crowd level and density is underlined in Table III, the results were showed when some motions and variation of the crowd happened.

As Table II shown, the estimation result (for same level) reach mean 92% CR, for Table III, the crowd level and density reach a mean of 81% CR when some crowds were moving. Although the effect of estimation is worse than the former, the difference is not too much to be unacceptable. It is because that the RSSI change in moving circumstance is more sensitive than those in static circumstance.

C. Impact of distribution

In this subsection, we explore how the crowd distribution affects estimation accuracy. There are three cases: case a, 2 crowds and each consist of 6 persons, they lie in both sides of the floor respectively; case b, 4 crowds and each consist of 3 persons, they lie in the four corner of the floor respectively; case c, 12 persons stay together and move random in the center of the floor. Figure 4 shows the impact of distribution. The estimation accuracy in case c is high than other two cases, the main reason is that the crowd is most similar to stochastic distribution in case c.

D. Impact of number of signal vector samples

Our approach detect crowd through a clustering process using the Wi-Fi signal vector samples gathered by person mobile phones. Therefore, the accuracy of our approach depends on the number of the collected Wi-Fi signal vector samples. Figure 5 shows the impact of number of signal vector samples. Given the study including 12 participants and 2 test scenes, it shows that, as the collected Wi-Fi samples increase,

TABLE II: COMPARISON OF REAL AND ESTIMATION WHEN STATIC

Crowd level and Density	SA	SB	SC	SD
SA	94%			
SB		91%		
SC			95%	
SD				89%

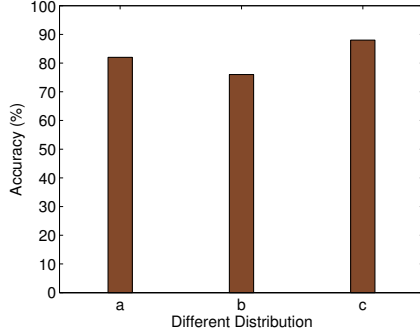


Fig. 4: Impact of crowd distribution

the accuracy of crowd detect also increase. For 200 or more samples, our approach achieves 92% accuracy, this shows that more training samples could make the result more accurate.

E. Impact of missing detection

We have described our method to detect the crowd and estimate density. However, there are false-negative cases that some devices may not be detected, especially at a crowd overlap and occlusion place. The mobile phone cannot handle them all within the limited time interval and Information processing ability while scanning for the surrounding mobile devices.

We take a set of experiments to verify the accuracy of our approach with some missing detected data, so we set a few of the mobile phones went into sleep or turn-off mode during the experiments. The results each mean 86% CR when the participants were stayed together. At the same time the approach achieve a mean of 73% CR when some crowds were moving. These results verify the reliability of our approach although the accuracy of monitoring have been a slight decline.

V. CONCLUSION

Crowd monitoring is an important service needed by many applications. Most of previous crowd monitoring approaches are video or image-based can only work in a line-of-sight pattern, and the privacy and security issue remain a concern. RF- based analysis solutions distributing the devices to each person is impractical and costly; attenuation model perform poor due to multipath effect; fingerprint methods need site survey, which is time-consuming, labor-intensive, and easily affected by environmental dynamics. This paper proposed a mobile phone based crowd detecting and density estimating approach, we using mobile phone Wi-Fi signal strength measurements clustering method to detect the crowd, the crowd density estimate by using Bluetooth detection. Extensive real-world experiments results demonstrate the feasibility and the

TABLE III: COMPARISON OF REAL AND ESTIMATION WHEN MOVING

Crowd level and Density	SA	SB	SC	SD
SA	85%			
SB		84%		
SC			82%	
SD				73%

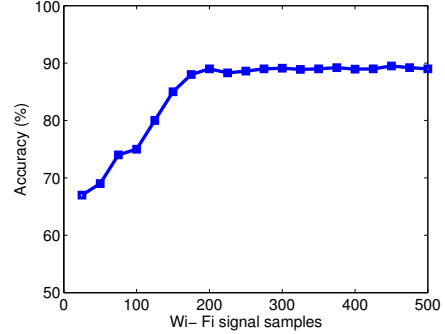


Fig. 5: Impact of number of signal vector samples

effectiveness of our design. For future work, In order to improve the crowd detection accuracy, we will utilize other sensor readings of the smart mobile phone to assist Wi-Fi feature clustering. On the other hand, we plan to perform detailed analysis on Bluetooth device detection reports, and employ clustering algorithm to estimate crowd density.

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